



E-Business

Assignment- Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 2 = 20

Question 1

Compared with bar codes, RFID improves automated data capture by enabling _____ identification of objects and reducing dependence on _____ during scanning.

- a) category-level, middleware
- b) product-family, network connectivity
- c) **unique-item, line-of-sight conditions**
- d) batch-level, frequency standards

Correct Option: c

Explanation: RFID supports unique item identification through EPC codes and can be read without direct line-of-sight.

Week 8 Slide: 5–6

Question 2

An RFID tag stores a unique identifier, the RFID _____ captures it, and the RFID _____ processes the captured data before updating enterprise systems.

- a) middleware, reader
- b) antenna, database
- c) database, middleware
- d) **reader, middleware**



Correct Option: d

Explanation: The reader captures tag information, while middleware processes and forwards it to backend systems.

Week 8 Slide: 9–10

Question 3

Two statements are given.

- I. Fixed RFID readers are generally installed at permanent locations.
- II. Multi-frequency readers can operate across more than one RFID frequency band.

Which option is correct?

- a) Only I is correct
- b) Only II is correct
- c) **Both I and II are correct**
- d) Neither I nor II is correct

Correct Option: c

Explanation: RFID readers may be classified as fixed/handheld and single-frequency/multi-frequency.

Week 8 Slide: 16

Question 4

Fill in the blanks.

Within the EPCglobal Network, the _____ resolves EPC references, while the _____ stores detailed information about the identified object.

- a) EPC Middleware, EPC Reader



b) ID System, Discovery Service

c) EPC Reader, EPC Middleware

d) **Discovery Services (ONS), EPC Information Services**

Correct Option: d

Explanation: ONS resolves EPC references and EPC Information Services maintain object-related information.

Week 8 Slide: 20, 25

Question 5

A manufacturer stores product information directly on rewritable RFID tags. Another company receiving the goods cannot interpret the stored information because the encoding scheme differs. This problem is primarily related to

a) electromagnetic interference

b) **lack of standardized data formats**

c) privacy protection

d) reader coordination

Correct Option: b

Explanation: RFID data representation on rewritable tags is not fully standardized.

Week 8 Slide: 33

Question 6

A shipment leaves a warehouse equipped with RFID infrastructure and travels across the country. During transit, managers still need continuous visibility of vehicle location. Which technology becomes essential after the shipment leaves RFID coverage?

a) EPC Information Service



b) RFID Middleware

c) **GPS**

d) Barcode scanning

Correct Option: c

Explanation: GPS is used once the vehicle moves beyond RFID reader coverage.

Week 8 Slide: 38

Question 7

A transportation planner wants software that can combine road maps, depot locations, customer positions, and vehicle attributes for route analysis. Which technology provides this capability?

a) RFID

b) GPS

c) EPC Network

d) **GIS**

Correct Option: d

Explanation: GIS integrates spatial and attribute data for analysis and decision-making.

Week 8 Slide: 45–48

Question 8

A logistics company must deliver goods using vehicles with limited carrying capacity. Each customer must be visited exactly once, and route demand cannot exceed vehicle capacity.

Which routing variant is most appropriate?

a) Vehicle Routing Problem with Time Windows (VRPTW)



b) **Capacitated Vehicle Routing Problem (CVRP)**

- c) Multiple Depot Vehicle Routing Problem (MDVRP)
- d) Periodic Vehicle Routing Problem (PVRP)

Correct Option: b

Explanation: CVRP incorporates vehicle-capacity constraints.

Week 8 Slide: 54

Question 9

Which situation best represents a Dynamic Vehicle Routing Problem (DVRP)?

- a) All customer requests are known before planning begins
- b) Routes remain fixed throughout execution
- c) Vehicle capacities are ignored during planning
- d) **New customer requests arrive after routes have already been generated**

Correct Option: d

Explanation: DVRP allows routing information to change after initial planning.

Week 8 Slide: 63–67

Question 10

Two statements are given.

- I. Tracking focuses on the current and future status of an item.
- II. Tracing focuses on reconstructing the historical path of an item.

Which option is correct?



- a) Only I is correct
- b) Only II is correct
- c) Neither I nor II is correct
- d) **Both I and II are correct**

Correct Option: d

Explanation: Tracking concerns present/future movement; tracing concerns past history.

Week 8 Slide: 79–80

*******END*******